FLAMeS had a higher lesion-wise F1 than three public CPU segmenters on thick-slice FLAIR in multiple sclerosis: a pre-registered paired benchmark

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Target journal: *NeuroImage: Clinical* (research article). **Alternative:** *Machine Learning for Biomedical Imaging (MELBA)*, which publishes reproducibility-oriented benchmarks.

Running title: CPU lesion segmenters vs FLAMeS on thick-slice FLAIR

Keywords: multiple sclerosis; lesion segmentation; FLAIR; slice thickness; benchmark; pre-registration; MindGlide; SAMSEG; WMH-SynthSeg; LST-AI; FLAMeS; nnU-Net

Highlights

- Pre-registered benchmark on 324 public MS scans, 179 with 3.9–8.3 mm FLAIR slices.
- FLAMeS had higher lesion-wise F1 than MindGlide, SAMSEG and WMH-SynthSeg.
- SAMSEG and WMH-SynthSeg had fewer false lesions but missed most small lesions.
- FLAMeS, trained only on voxels of 1.25 mm or less, led on 3.9–8.3 mm FLAIR.
- Every number traces to committed files; incremental CPU cost was 12.55 USD.

Abstract

Background. Clinical archives hold many 2D FLAIR scans with 4–8 mm slices, but most multiple sclerosis (MS) lesion segmenters are developed on 1 mm 3D FLAIR. Public CPU tools designed for heterogeneous resolution have not been compared with a FLAIR-only nnU-Net ensemble (FLAMeS) on such scans.

Methods. Two-sided paired comparisons and a reading rule for "outperforms" were registered in a public repository (2026-10-21, commit 65a95e3) before any tool output existed. Five public cohorts with expert masks: 179 thick-slice scans (3.9–8.3 mm; 160 patients) and 145 scans on a 1 mm grid (105 patients). MindGlide, WMH-SynthSeg and SAMSEG ran with default settings on CPU on the skull-stripped FLAIR used by the FLAMeS 5-fold ensemble; existing LST-AI maps were added on the 1 mm grid. Co-primary outcomes: lesion-wise F1 and false lesions per scan; cohort-stratified patient-cluster bootstrap with Holm correction.

Results. On thick slices FLAMeS had a higher lesion-wise F1 (0.724) than every tool: ΔF1 −0.170 (95% CI −0.206 to −0.136) for MindGlide, −0.237 (−0.304 to −0.173) for SAMSEG (129 scans) and −0.384 (−0.424 to −0.344) for WMH-SynthSeg (all $p_{Holm} < 0.002$). SAMSEG and WMH-SynthSeg had 5.15 and 3.88 fewer false lesions per scan (p_{Holm}

< 0.002) but detected only 6% and 5% of expert lesions of 14–50 mm^3 (FLAMeS 49%); MindGlide's false-lesion difference was not significant (+0.57, −0.89 to +2.21). No tool met the pre-specified rule for outperforming FLAMeS, and none was equivalent. On the 1 mm grid FLAMeS also led (LST-AI −0.180, −0.212 to −0.150). Incremental compute cost: 12.55 USD.

Conclusions. In pre-registered paired comparisons on real thick-slice FLAIR, FLAMeS had a higher lesion-wise F1 than every public CPU tool tested, and none met the pre-specified criterion for outperforming it. Fewer false lesions from conservative tools came at a large cost in sensitivity. No clinical utility is claimed.

1. Introduction

T2-weighted fluid-attenuated inversion recovery (FLAIR) lesions are central to the diagnosis and monitoring of multiple sclerosis (MS) (Thompson et al. 2018; Filippi et al. 2019). The 2021 MAGNIMS–CMSC–NAIMS recommendations make 3D FLAIR the core brain sequence and accept high-quality 2D acquisitions with slices of at most 3 mm and no gap as an alternative (Wattjes et al. 2021). Many clinical archives predate or do not follow these recommendations: the three public clinical cohorts used here, for example, contain axial 2D FLAIR with 3.9–8.3 mm between slice centres. The developers of recent segmentation tools note that high-resolution 3D scans are still not always acquired in routine practice (Dereskewicz, La Rosa, Dos Santos Silva, et al. 2025) and have built tools to reuse such archives (Goebl et al. 2025). Slice thickness affects both how many MS lesions are detected and how their volume is measured (Molyneux et al. 1998). Partial-volume averaging across thick slices blurs small lesions into the surrounding tissue, so tools trained on 1 mm data meet both a resolution shift and a change in lesion appearance.

Most automated MS lesion segmenters were developed and challenged on 3D or near-isotropic research data, such as the ISBI-2015 and MSSEG-2016 challenges (Carass et al. 2017; Commowick et al. 2018, 2021); their clinical integration and the evidence for commercial tools have been reviewed (Spagnolo et al. 2023; Mendelsohn et al. 2023), and distribution shift between sites is a recognised challenge (Malinin et al. 2022). Three families of public tools address heterogeneous resolution explicitly. Domain-randomised networks are trained on synthetic images of random contrast and resolution: SynthSeg and SynthSR for anatomy (Billot et al. 2023; Iglesias et al. 2023), WMH-SynthSeg for white-matter hyperintensities (Laso et al. 2024), and MindGlide, which was trained on real archive scans plus synthetic scans with resolutions of 1–9 mm (Goebl et al. 2025). Generative, contrast-adaptive models such as SAMSEG fit an atlas and a lesion shape model to each scan (Cerri et al. 2021). Supervised ensembles such as LST-AI (FLAIR plus T1) (Wiltgen et al. 2024) and FLAMeS (FLAIR only; nnU-Net (Isensee et al. 2021)) are trained on labelled real scans (Dereskewicz, La Rosa, Dos Santos Silva, et al. 2025). All of them run on a CPU, which matters in hospitals without graphics processors.

Our prior expectation, which was not part of the protocol, was that a tool built for heterogeneous resolution might do better on thick slices than a model trained on high-resolution data. To our knowledge this has not been tested on real thick-slice 2D FLAIR with expert masks from several sources. Tool comparisons are often run by a tool's developers, on data that may overlap with training sets, with analysis choices made after the results are seen (Varoquaux and Cheplygina 2022; Kapoor and Narayanan 2023). Pre-registration fixes the questions, metrics and tests before outcomes exist, so that a result is informative whichever way it falls (Nosek et al. 2018).

We therefore pre-registered a paired benchmark of three public CPU tools — MindGlide, WMH-SynthSeg and SAMSEG — against the FLAMeS 5-fold ensemble on 179 real thick-slice FLAIR scans from three public cohorts, with LST-AI added on scans with a 1 mm grid. The primary question was two-sided: does any tool achieve a different lesion-wise F1, or a different number of false lesions per scan, than FLAMeS on thick slices? A pre-stated reading rule defined when a tool "outperforms" FLAMeS. Secondary questions were the same comparison on the 1 mm grid, a between-band contrast of each tool's difference from FLAMeS (registered as a "degradation contrast"), and whether

giving MindGlide the non-skull-stripped FLAIR it was designed for changes its results. An exploratory question asked whether disagreement with a second tool flags scans on which FLAMeS performs poorly. FLAMeS had the higher lesion-wise F1 in every primary comparison, and no tool met the reading rule.

2. Methods

2.1 Design, pre-registration and reporting

This is a retrospective, paired, multi-cohort benchmark on public de-identified data. The protocol (version 1.0, 2026-10-21), the metric code (`research/paper_f/metrics.py`), the evaluation code (`research/paper_f/evaluate.py`) and the frozen list of input files with their SHA-256 hashes (`research/paper_f/flames_inputs.json`) were recorded in commit `65a95e310168071703398aab3a53ba9fc0086e1a` (2026-10-21 20:45 UTC). The branch containing it was first pushed to the public GitHub repository `nicolasbonilla/medical-imaging-viewer` at 2026-10-21 20:58 UTC (GitHub repository activity log), 40 minutes before the first tool container started (21:39 UTC; the first virtual machine had been created at 21:31 UTC). The SHA-256 of the protocol file as stored at that commit is `587ba9ffc1f07d559a0008728e97ec9e1e9455eaf0037e37bbcc52b8f5f284e2`. Every later change was logged with its date, reason and what had been seen (Section 2.10, Table 6). The pre-registration is time-stamped by git and GitHub; it was not lodged with an independent registry, and the public visibility of the branch on 2026-10-21 rests on GitHub's activity log [TODO owner: archive the protocol as of commit 65a95e3 on OSF or Zenodo with a DOI, and cite it as a retrospective archive of a time-stamped protocol]. The protocol registered two-sided questions and a reading rule; it stated no directional hypothesis. Reporting was guided by the CLAIM 2024 checklist (Tejani et al. 2024) [TODO owner: complete and attach the checklist]. The final analysis ran once on the full outputs from commit `7a9a24e8fa92c83ded210854fd8614f7be4dcd50` (2026-10-29 00:11 UTC; pushed 00:01 UTC), which records the SHA-256 of all 3,059 input files read; an earlier, unplanned interim analysis is described in Section 2.10.

Roles in the research process, including an AI coding agent. The author chose the research question and set the constraints (CPU only, a budget cap, no clinical claims). An AI coding agent (Claude Opus 5.5, Anthropic, used through Claude Code) drafted the protocol and the runbook, wrote all analysis and cloud-job code and its tests, and ran the analyses. Acting as the run coordinator, the agent also took run-time decisions: it applied the runbook's memory rule that led to the WMH-SynthSeg `--crop` deviation and applied the pre-stated budget rule that selected the SAMSEG scan set (Table 6). It verified the references and wrote the first draft of this manuscript. All 13 commits of the study branch carry a `Co-Authored-By` trailer naming the agent. [Owner to confirm: whether the author reviewed and approved the protocol and each amendment before the runs, and who launched each cloud run. The repository holds no record of these approvals, and this manuscript does not assert them.]

2.2 Cohorts

Five public cohorts with expert lesion masks were fixed in the protocol (Table 1): 324 scans of 265 patients. The inputs were the exact FLAIR files on which the FLAMeS maps had been computed; masks were on the same grid.

1 mm-grid band ("iso"; 145 scans, 105 patients). MSLesSeg, the dataset of the ICPR 2024 challenge (Guarnera et al. 2025b, 2025a; Rondinella et al. 2025), provides 115 longitudinal scans of 75 patients (1.5 T, Catania) as skull-stripped images on a 1 mm MNI grid; masks were drawn by one rater and validated by a neuroradiologist and a neurologist. Its native FLAIR is not 1 mm: the data paper reports a mean slice thickness of 2.61 mm (SD 0.62), and the images were registered with FLIRT to the 1 mm MNI152 template (Guarnera et al. 2025b). Ljubljana (Lesjak et al. 2018) provides 30 patients scanned with near-isotropic 3D FLAIR at 3 T (0.47 × 0.47 × 0.8–0.9 mm voxels), with a multi-rater consensus mask, also used on the 1 mm MNI grid. As the protocol (Section 6) anticipated, "iso" therefore

describes the grid the tools received, not the acquisition: 115 of its 125 FLAMeS comparison scans were acquired with slices of about 2.6 mm.

Thick band (> 3 mm; 179 scans, 160 patients). MS3SEG (Bawil et al. 2026; Bashiri Bawil 2026): 100 patients, one 1.5 T Toshiba scanner, axial 2D FLAIR, 0.898 × 0.898 mm in plane, 6.0–8.1 mm slice spacing (median 7.2 mm). Its MS-lesion mask excludes periventricular hyperintensities that the annotators judged not to be MS lesions. Muslim et al. (Muslim et al. 2022; Muslim 2022): 51 axial scans (of 60; 9 coronal or sagittal scans were excluded by a pre-stated rule), 1.5 T, about 20 centres, 4.86–8.3 mm slice spacing (median 6.0 mm). The files carry no geometry, so the in-plane pixel size (0.41–0.97 mm) was estimated from head dimensions calibrated on MS3SEG (expected error about ±5% per axis); left and right cannot be recovered. PediMS (Popa et al. 2025b, 2025a): 28 longitudinal scans of 9 children, 3 T Siemens Skyra, 0.75 × 0.75 × 3.9 mm; one child has Baló's concentric sclerosis.

Skull stripping: MSLesSeg and Ljubljana were distributed skull-stripped; PediMS ships SynthStrip brains (Hoopes et al. 2022); MS3SEG and Muslim (151 of the 179 thick scans) were stripped with HD-BET 2.0.1 on CPU without test-time augmentation (Isensee et al. 2019), whose model parameters are licensed for non-commercial use only (CC BY-NC 4.0; Table 7). Bands were defined by the largest voxel dimension of the grid the tools received: iso ≤ 1.5 mm and thick > 3 mm, with exploratory sub-bands 3–5.5 mm (PediMS and 24 Muslim scans) and > 5.5 mm (MS3SEG and 27 Muslim scans).

Table 1. Cohorts. Slice spacing from the headers of the files the tools received (`data/extra_numbers.json`); expert lesions counted after the 14.1 mm^3 size filter.

Cohort (licence)	Scans / patients	Field, acquisition	Grid given to the tools	Skull stripping	Reference standard
MSLesSeg (CC BY 4.0)	115 / 75	1.5 T, one city, longitudinal; native slices ~2.6 mm	MNI 1 mm (resampled)	provided	one rater, validated by neuroradiologist and neurologist
Ljubljana (CC BY-NC-ND per the lab; CC BY per the mirror)	30 / 30	3 T Siemens, 3D FLAIR	MNI 1 mm	provided	three-rater consensus
MS3SEG (CC BY 4.0)	100 / 100	1.5 T Toshiba, axial 2D	0.898^2 × 6.0–8.1 mm	HD-BET 2.0.1	junior annotator and senior neuroradiologist; "normal" WMH excluded
Muslim et al. (CC BY 4.0)	51 / 51	1.5 T, ~20 centres, axial 2D	$0.41–0.97^2$ (estimated) × 4.86–8.3 mm	HD-BET 2.0.1	resident → consultant radiologist → neurologist
PediMS (CC BY 4.0)	28 / 9	3 T Siemens, 2D, children	0.75^2 × 3.9 mm	SynthStrip (dataset)	one rater, validated by paediatric neurologist and radiologist

In the isotropic band the FLAMeS comparison set contained 3,463 expert lesions, 1,209 (34.9%) of them 14–50 mm^3; in the thick band 5,993 expert lesions, 1,248 (20.8%) of them 14–50 mm^3. No scan had an empty reference.

2.3 Tools

Table 2 lists the tools; full container digests and weight hashes are in Appendix Table A1. **FLAMeS** is an nnU-Net v2 (Isensee et al. 2021) FLAIR-only model released as five cross-validation folds (Zenodo 17955359, version 2.0, CC

BY 4.0) (La Rosa 2025; Dereskewicz, La Rosa, Dos Santos Silva, et al. 2025; Dereskewicz, La Rosa, Santos Silva, et al. 2025; Journal of Neuroimaging 2025). Its maps had been computed for earlier studies on the same inputs (4 vCPU, tile step 0.5 with Gaussian weighting, PyTorch 2.14.1 CPU build; nnU-Net package version frozen in the job's lock file but not recorded in the run manifests [TODO owner: read it from the lock file]); the prediction was the mean of the five fold probabilities thresholded at 0.5. Test-time mirroring, nnU-Net's default, was switched off to save CPU time, whereas the comparators ran at their defaults; mirroring usually helps slightly, so this choice is unlikely to have favoured FLAMeS. **LST-AI** 1.2.0 (three U-Nets, FLAIR + T1) (Wiltgen et al. 2024) had been run on the 1 mm cohorts only (144 scans; one MSLesSeg scan has a blank T1); its ensemble probability was thresholded at 0.5, LST-AI's own binarisation. **MindGlide** 1.3.0 (Goebl et al. 2025) returns a label map on the input grid (lesion = label 18). **WMH-SynthSeg** (Laso et al. 2024) and **SAMSEG** with its lesion model (Cerri et al. 2021) were run from FreeSurfer 8.2.0 (Fischl 2012) (labels 77 and 99). LST-AI was not run on the thick cohorts because it needs a co-registered T1 that was not prepared; tools needing a GPU, MATLAB or per-site training were excluded in the protocol.

Table 2. Tools, training data and overlap with the five cohorts.

Tool	Version	Input; lesion output	Training data (as published)	Overlap with our cohorts
FLAMeS (reference)	weights v2.0 (2025-12-16), nnU-Net v2, 5 folds	FLAIR; mean fold probability > 0.5	668 FLAIR scans of 575 people with MS from seven sites, including ISBI-2015 (20 scans), MSSEG-2016 (53) and Ljubljana (20); every voxel dimension in the released fingerprint ≤ 1.25 mm	20 Ljubljana scans in training → excluded from every FLAMeS comparison; MSLesSeg was one of its external test sets
LST-AI	1.2.0 (Docker digest, Table A1)	FLAIR + T1; probability > 0.5	491 in-house 3D FLAIR + T1 scans, one centre	none; Ljubljana was one of its published test sets
MindGlide	1.3.0 (container and weights hash, Table A1)	FLAIR; label 18	4,247 real scans from clinical-trial and academic archives + 4,303 synthetic scans of random 1–9 mm resolution; labels from an automated method	none; Ljubljana and ISBI-2015 in its external test
WMH-SynthSeg	FreeSurfer 8.2.0; model v10 (231110)	FLAIR; label 77 (WMH)	synthetic images generated from label maps (including ISBI-2015 and MSSEG-2016), random slice spacing 2.5–8.5 mm	none
SAMSEG `--lesion`	FreeSurfer 8.2.0	FLAIR; label 99	atlas from 20 subjects; lesion shape model from 212 MS subjects	none

Training overlap with FLAMeS (details in Appendix B). FLAMeS was trained on 20 of the 30 Ljubljana scans; we identified them exactly from the dataset fingerprint shipped with the weights (patients 01–15 and 21–25) and excluded them from every comparison with FLAMeS (they appear only in sensitivity analysis S5). MSLesSeg was not a FLAMeS training source but one of its external test sets, and no FLAMeS training case has a voxel dimension above 1.25 mm, so no thick-slice scan was a training case at its own geometry.

2.4 Processing

Every tool received the byte-identical skull-stripped FLAIR used for FLAMeS (SHA-256 checked on each machine against the frozen list; no mismatch occurred), with default settings and only the flags below, on 4 vCPUs, inside containers pinned by digest and run without network access. MindGlide: `--device cpu`. WMH-SynthSeg: `mri_WMHsynthseg--devicecpu--threads4--save_lesion_probabilities`. SAMSEG: `run_samseg--lesion--lesion-mask-pattern1--pallidum-separate--threads4--random-seed12345--save-posteriorsLesions`. MindGlide and SAMSEG return the input grid (identity transfer). WMH-SynthSeg returns a 1 mm grid; for thick scans each FLAMeS voxel received the mean lesion fraction over its footprint (sampled at ⌈edge length in source voxels⌉ points per axis) and was labelled lesion when the fraction was ≥ 0.5 (179 scans); 1 mm outputs were mapped trilinearly (145 scans). This majority rule, fixed in the protocol, is the only resampling applied to any tool's output; because it can remove thin lesions from a thick voxel, a post hoc sensitivity analysis uses WMH-SynthSeg's saved lesion probabilities instead (Section 3.4). A run that failed or exceeded 45 minutes was to be scored as an empty prediction; no scan ended that way.

2.5 Outcomes

Definitions were fixed in code before any output existed. A lesion is a 3D connected component with 26-connectivity on the FLAMeS grid. Components smaller than 14.137 mm^3 (a sphere of 3 mm diameter) were removed from prediction and reference before matching. A predicted component is a true positive if it overlaps any reference component by at least one voxel; a reference lesion is detected if any predicted component overlaps it. Lesion-wise precision = matched predicted / predicted; recall (sensitivity) = detected reference / reference; F1 = 2PR/(P+R), set to 0 when nothing is predicted. **Co-primary outcomes:** lesion-wise F1 and false lesions per scan (predicted components touching no reference lesion). **Secondary outcomes:** voxel-wise Dice on the unfiltered masks (Dice 1945), lesion-wise precision and recall, detection by reference lesion size (14–50, 50–200, 200–1000 and ≥ 1000 mm^3), predicted and reference lesion volume, failure rate, CPU time and peak memory. The choice of instance-level metrics is consistent with current recommendations for detection-type tasks (Maier-Hein et al. 2024).

2.6 Statistical analysis

For every comparison we computed per-scan differences (tool − FLAMeS) on scans where both outputs were usable and neither was contaminated. The estimate was the mean over cohorts of the per-cohort mean difference (each cohort counts once). Uncertainty came from a patient-cluster bootstrap stratified by cohort (Field and Welsh 2007; Efron and Tibshirani 1993): patients were resampled with replacement within each cohort, 10,000 replicates, seed 20261021, percentile 95% intervals. The two-sided bootstrap p was twice the smaller tail proportion of replicates at or beyond zero. When no replicate fell on the other side of zero, p is below the resolution of the bootstrap (2/10,000); we report such Holm-adjusted values as $p_{Holm} < 0.002$ (families of 4 or 6 tests) or < 0.003 (family of 11), rather than the "< 0.0001" printed by the analysis tables. With 10,000 replicates the Monte Carlo standard error of a bootstrap p near 0.45 is about 0.008, so other p-values are given to two significant figures. Equivalence was declared when the 90% interval lay inside ±0.05 F1 or ±1 false lesion per scan, the interval form of two one-sided tests (Schuirmann 1987). The protocol did not justify these margins; for scale, 0.05 is 6.9% of FLAMeS's thick-band F1 and one false lesion is 14% of its 6.9 false lesions per scan.

Primary family (6 tests, Holm (Holm 1979), α = 0.05): ΔF1 and Δ false lesions per scan for MindGlide, WMH-SynthSeg and SAMSEG vs FLAMeS on the thick band. **Reading rule:** a tool *outperforms* FLAMeS only if $p_{Holm} < 0.05$ with ΔF1 > 0 and Δ false lesions not worse than +1 per scan, or $p_{Holm} < 0.05$ with fewer false lesions and ΔF1 not below −0.05. **Secondary family (11 tests, Holm):** the same two outcomes on the iso band for the three tools and LST-AI (8 tests), and, for the three tools, the contrast registered as the "degradation contrast": the tool's paired ΔF1 on the

thick band minus its paired ΔF1 on the iso band (3 tests). The two bands share no patient, so their bootstraps are independent; the contrast compares different cohorts, acquired differently, and is therefore not a measure of slice thickness alone. **Raw-input arm (4 tests, Holm; registered before any full-run output was examined, Section 2.10):** MindGlide on the FLAIR before skull stripping vs FLAMeS, and vs MindGlide on the stripped FLAIR, on the thick band. Tests that could not be run are reported as "not run" and enter the Holm procedure over the registered family size as p = 1 (protocol Sections 4.4 and 12); a tool not run on a cell is never imputed.

Sensitivity analyses (primary family, no adjustment): S1 no size filter; S2 3 mm^3 filter; S3 in-plane 8-connectivity (components do not join across thick slices); S4 complete cases; S5 iso comparisons including the 20 FLAMeS training scans; S6 thick band without Muslim; S7 scan-weighted estimate. **Exploratory (labelled as such):** E1, the area under the ROC curve (Hanley and McNeil 1982) of tool–FLAMeS disagreement (1 − lesion-wise F1 between the two outputs) for identifying scans with FLAMeS lesion-wise F1 < 0.5; E2, keeping only FLAMeS lesions that the tool also marks. **Post hoc additions** (not in the protocol; labelled wherever used; scripts `compute_extra_numbers.py`, `wmh_resampling_posthoc.py`, `fingerprint_check.py`): an interval, a cohort-stratified version and FLAMeS-only baselines for E1; precision, recall and Dice per band; paired-set, per-cohort and scan-pooled (micro-averaged) analyses; the WMH-SynthSeg resampling analysis, whose variants were fixed before any was computed; and the fingerprint check of Section 2.3.

Precision and weighting. With a per-scan SD of 0.15 for ΔF1, the protocol anticipated a 95% half-width of about ±0.04 F1 on the thick band. Because each cohort counts once, the 9 PediMS children weigh as much as the 100 MS3SEG patients, and on the iso band the 10 FLAMeS-clean Ljubljana patients carry half of every estimate against 115 MSLesSeg scans. Software: Python 3.11.4 and NumPy 1.26.4.

2.7 SAMSEG analysis set: pre-stated budget tiers

The protocol fixed a run order (cheapest tool first) and budget tiers that depended only on measured runtime, never on accuracy, so that the projected total spend stayed ≤ 15 USD: tier A all 324 scans, tier B the 179 thick scans, tier C a fixed thick subsample of 129 scans (all PediMS, all Muslim and the 50 odd-numbered MS3SEG scans). Applied after the other tools had run, with the real spend so far, the rule selected tier C (projected total 13.76 USD). SAMSEG comparisons are therefore paired on 129 thick scans of 110 patients, and the three SAMSEG tests of the secondary family were not run.

2.8 Compute environment and cost

All tool runs used Google Cloud virtual machines with 4 vCPUs (n2d-standard-4, n2-standard-4, e2-standard-4; n2d-highmem-4 with 32 GB for WMH-SynthSeg) in region us-central1. Costs are list prices recorded per run in a compute ledger (snapshot `data/compute_ledger_c8b4239.csv`). The study's own runs used 14 virtual machines, 61.90 machine-hours and 12.55 USD (incremental cost; Table 5). The FLAMeS, HD-BET and LST-AI maps came from earlier runs on the same inputs; including them, the total is 101.97 machine-hours and 19.84 USD at list price (list prices are not yet reconciled with billing).

2.9 Integrity checks and blinding

Before the analysis, a script checked every planned output against the run manifests (presence, SHA-256, failures; `results/verify_runs.json`). A structural check (`evaluate.py check`) then confirmed that every mask lay on the exact FLAMeS grid and was binary. That check also computes, for every output, the Dice, lesion-wise F1 and predicted volume against the expert mask, and flags a Dice below 0.1 as a probable label or orientation error (`results/structural_check_wmhsynthseg.json` and `results/structural_check_samseg.json`; `results/RESULTS.md` quotes

median Dice values from them). For WMH-SynthSeg and SAMSEG it was run on 2026-10-29 at 00:01:59 UTC, 21 s after the decisions of that date were pushed (00:01:38 UTC) and nine minutes before the final analysis (00:11 UTC). The repository does not prove this order: the timestamp comes from the agent's session log, which is kept outside the repository, and the check's output files carry no time stamp [TODO owner: decide whether to archive the relevant session-log excerpt]. Otherwise, for each tool the only accuracy numbers looked at before its full run were pilot plausibility checks on three scans; the interim analysis of Section 2.10 is the exception.

2.10 Deviations from the protocol

All deviations are listed in Table 6 (times in UTC). In short: a secondary arm with MindGlide on the raw FLAIR was registered before any full-run output was examined (the MindGlide full-run machines had started about ten minutes earlier, so some outputs probably existed); WMH-SynthSeg ran without the pre-stated `--crop` memory fallback (one out-of-memory scan was re-run with it); the budget rule gave SAMSEG tier C; Holm was applied over the registered family size with tests not run entered as p = 1; and an interim analysis of MindGlide and LST-AI was run on 2026-10-22, although the protocol stated that the analysis would run once on the full outputs. The interim included the exploratory E1 AUROC of 0.78, which the product-planning note (`docs/research/paper-f/APP-FEATURE.md`) then read as meeting its criterion for a "second-opinion" feature. That interim look was not logged as a deviation at the time; it was identified while writing this report. No outcome definition, band, threshold or test changed.

3. Results

3.1 Completeness

All planned outputs were present — WMH-SynthSeg 324/324, MindGlide 324/324, MindGlide on raw input 179/179 and SAMSEG 129/129 — and all but one matched the hash recorded on the machine (Table A2). One WMH-SynthSeg run (muslim_16) stopped for lack of memory and was re-run with `--crop`, as pre-stated. The three MindGlide pilot scans failed in the first pilot runs because the container could not write to its output folder, and were re-run after the fix. One SAMSEG output (ms3seg_001, pilot) has no manifest record because the pilot's manifest write crashed; the file was verified by its lesion-voxel count (1,129) and kept. No scan was scored as a failure, no input hash mismatched, and every mask passed the structural check. SAMSEG predicted no lesion after the size filter in 3 of 129 scans; every other tool predicted at least one lesion in every scan.

3.2 Primary comparisons: thick-slice FLAIR

FLAMeS had a cohort-weighted lesion-wise F1 of 0.724 (95% CI 0.702–0.745) and 6.9 false lesions per scan on the 179 thick scans (Table 4). All three CPU tools had a lower lesion-wise F1 than FLAMeS (Table 3, Figure 1): MindGlide by 0.170 (95% CI 0.136–0.206), SAMSEG by 0.237 (0.173–0.304) and WMH-SynthSeg by 0.384 (0.344–0.424); p_{Holm} < 0.002 for each. SAMSEG had 5.15 fewer false lesions per scan (95% CI 3.91–6.30) and WMH-SynthSeg 3.88 fewer (2.27–5.21), both p_{Holm} < 0.002. For MindGlide the difference in false lesions was +0.57 per scan (−0.89 to +2.21; p_{Holm} = 0.45): no difference was shown and equivalence within ±1 was not shown either. Of the six primary tests, five rejected the null hypothesis of no difference: the three F1 tests in favour of FLAMeS and the two false-lesion tests in favour of SAMSEG and WMH-SynthSeg. Under the pre-stated reading rule no tool outperformed FLAMeS, because every ΔF1 was far below −0.05, and no tool was equivalent to FLAMeS. On the 129 SAMSEG scans, FLAMeS had F1 0.729 and 6.97 false lesions per scan, against 0.492 and 1.82 for SAMSEG. The direction held in every cohort: on the thick cohorts the per-cohort ΔF1 ranged from −0.123 to −0.226 for MindGlide, from −0.226 to −0.255 for SAMSEG and from −0.279 to −0.467 for WMH-SynthSeg, every 95% interval excluding zero (post hoc; Table A3).

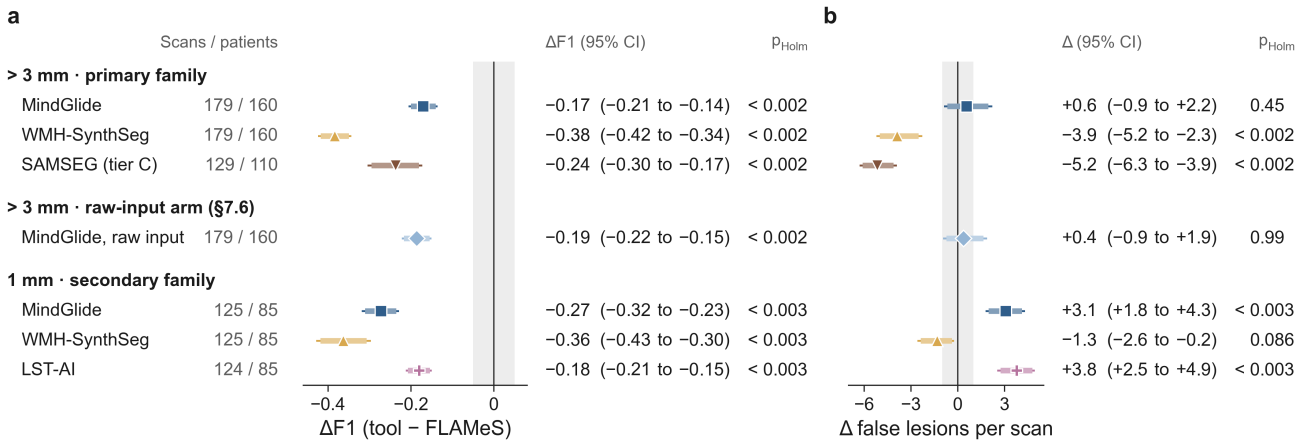


Figure 1. Paired differences from FLAMeS (tool − FLAMeS) in lesion-wise F1 (a) and false lesions per scan (b); the estimate and its 95% CI are printed beside each row. Rows give scans / patients. Shaded bands are the pre-registered equivalence margins (±0.05 F1; ±1 false lesion per scan). Thin lines are 95% and thick, lighter lines 90% patient-cluster bootstrap intervals (the 90% interval is the equivalence criterion), from 10,000 replicates stratified by cohort. p-values are Holm-adjusted over the registered family size, with tests not run counted as p = 1; '< 0.002' and '< 0.003' mean that no replicate crossed zero. Each tool keeps one colour and marker in every figure; FLAMeS, the reference, is dark grey.

Table 3. Pre-registered comparisons (difference = tool − reference; patient-cluster bootstrap stratified by cohort; `results/comparisons.json`).

Family	Comparison	Band	Scans / patients	ΔF1 (95% CI)	p_{Holm}	Δ false lesions / scan (95% CI)	p_{Holm}
Primary (6)	MindGlide – FLAMeS	thick	179 / 160	−0.170 (−0.206 to −0.136)	< 0.002	+0.57 (−0.89 to +2.21)	0.45
Primary (6)	WMH-SynthSeg – FLAMeS	thick	179 / 160	−0.384 (−0.424 to −0.344)	< 0.002	−3.88 (−5.21 to −2.27)	< 0.002
Primary (6)	SAMSEG – FLAMeS	thick	129 / 110	−0.237 (−0.304 to −0.173)	< 0.002	−5.15 (−6.30 to −3.91)	< 0.002
Secondary (11)	MindGlide – FLAMeS	iso	125 / 85	−0.272 (−0.319 to −0.229)	< 0.003	+3.07 (+1.78 to +4.32)	< 0.003
Secondary (11)	WMH-SynthSeg – FLAMeS	iso	125 / 85	−0.363 (−0.429 to −0.297)	< 0.003	−1.32 (−2.58 to −0.23)	0.086
Secondary (11)	LST-AI – FLAMeS	iso	124 / 85	−0.180 (−0.212 to −0.150)	< 0.003	+3.79 (+2.53 to +4.95)	< 0.003
Secondary (11)	SAMSEG – FLAMeS	iso	—	not run (tier C)[b]	—	not run[b]	—
Secondary (11)	Between-band contrast[c], MindGlide	thick − iso	304 / 245	+0.102 (+0.047 to +0.160)	< 0.003	—	—

Family	Comparison	Band	Scans / patients	ΔF1 (95% CI)	p_{Holm}	Δ false lesions / scan (95% CI)	p_{Holm}
Secondary (11)	Between-band contrast[c], WMH-SynthSeg	thick – iso	304 / 245	−0.021 (−0.098 to +0.056)	1.0	—	—
Secondary (11)	Between-band contrast[c], SAMSEG	thick – iso	—	not run[b]	—	—	—
Raw arm (4)	MindGlide raw – FLAMeS	thick	179 / 160	−0.186 (−0.222 to −0.150)	< 0.002	+0.38 (−0.94 to +1.88)	0.99
Raw arm (4)	MindGlide raw – MindGlide	thick	179 / 160	−0.016 (−0.028 to −0.004)	0.037	−0.19 (−0.75 to +0.32)	0.99

[b] Not run: SAMSEG was run only on its thick tier-C scans. A test not run enters Holm over the registered family size as p = 1 (a conservative convention registered before any WMH-SynthSeg or SAMSEG accuracy was computed); it changed one conclusion, the WMH-SynthSeg false-lesion test on the iso band (Holm over the eight tests performed: 0.034). [c] Registered as the "degradation contrast": the tool's paired ΔF1 on the thick band minus that on the iso band; the bands are different cohorts (Section 3.5).

Table 4. Absolute performance per band (cohort-weighted mean, 95% CI). F1, false lesions and Dice from `results/tables.md` (Table 2); precision and recall descriptive (`data/extra_numbers.json`). The rows are not all on the same scans (see the notes); paired comparisons are in Table 3.

Band	Tool	Scans	Lesion-wise F1	False lesions / scan	Precision	Recall	Dice
iso	FLAMeS	125	0.765 (0.727–0.798)	3.8 (2.8–5.0)	0.819	0.740	0.657
iso	LST-AI	144	0.620 (0.588–0.650)	8.4 (7.2–9.8)	0.690	0.589	0.520
iso	MindGlide	145	0.518 (0.490–0.543)	6.3 (5.6–7.0)	0.569	0.599	0.418
iso	WMH-SynthSeg	145	0.387 (0.354–0.419)	2.3 (1.9–2.6)	0.752	0.312	0.306
thick	FLAMeS	179	0.724 (0.702–0.745)	6.9 (6.0–7.8)	0.738	0.758	0.642
thick	MindGlide	179	0.554 (0.517–0.589)	7.5 (6.7–8.5)	0.600	0.593	0.493
thick	MindGlide, raw input	179	0.538 (0.503–0.572)	7.3 (6.6–8.0)	0.606	0.549	0.508
thick	WMH-SynthSeg	179	0.341 (0.308–0.373)	3.0 (2.4–3.9)	0.707	0.248	0.250
thick	SAMSEG (tier C)	129	0.492 (0.427–0.553)	1.8 (1.4–2.3)	0.837[a]	0.397	0.479

[a] 126 scans; precision is undefined when no lesion is predicted. On the same tier-C scans FLAMeS's precision was 0.743. Iso rows for the tools include all 30 Ljubljana scans; FLAMeS includes only the 10 it was not trained on.

3.3 Why the conservative tools had fewer false lesions

WMH-SynthSeg and SAMSEG marked much less lesion than the experts (Figure 2). On thick scans the mean predicted lesion volume per cohort was 4.2–5.2 mL for WMH-SynthSeg and 6.8–9.7 mL for SAMSEG, against expert volumes of 12.6–19.8 mL and FLAMeS volumes of 13.1–18.4 mL (`results/tables.md`, Table 1). Their lesion-wise precision was higher than or similar to that of FLAMeS (SAMSEG 0.837 against 0.743 for FLAMeS on the same 129 scans; WMH-SynthSeg 0.707 against 0.738), but their recall was much lower (0.397 and 0.248 vs 0.758). The deficit concentrated in small lesions (Figure 3): of the 1,248 expert lesions of 14–50 mm^3 on thick scans, FLAMeS detected 0.49 (95% CI 0.44–0.54), MindGlide 0.22 (0.18–0.27), SAMSEG 0.06 (0.04–0.09; 935 lesions in its scans) and WMH-SynthSeg 0.05 (0.03–0.07). For lesions ≥ 1000 mm^3, FLAMeS and MindGlide both detected 0.97, SAMSEG 0.89 and WMH-SynthSeg 0.83. The fewer false lesions of these two tools thus reflect a more conservative operating point; they are not evidence of a better trade-off, because sensitivity fell far more than false lesions did.

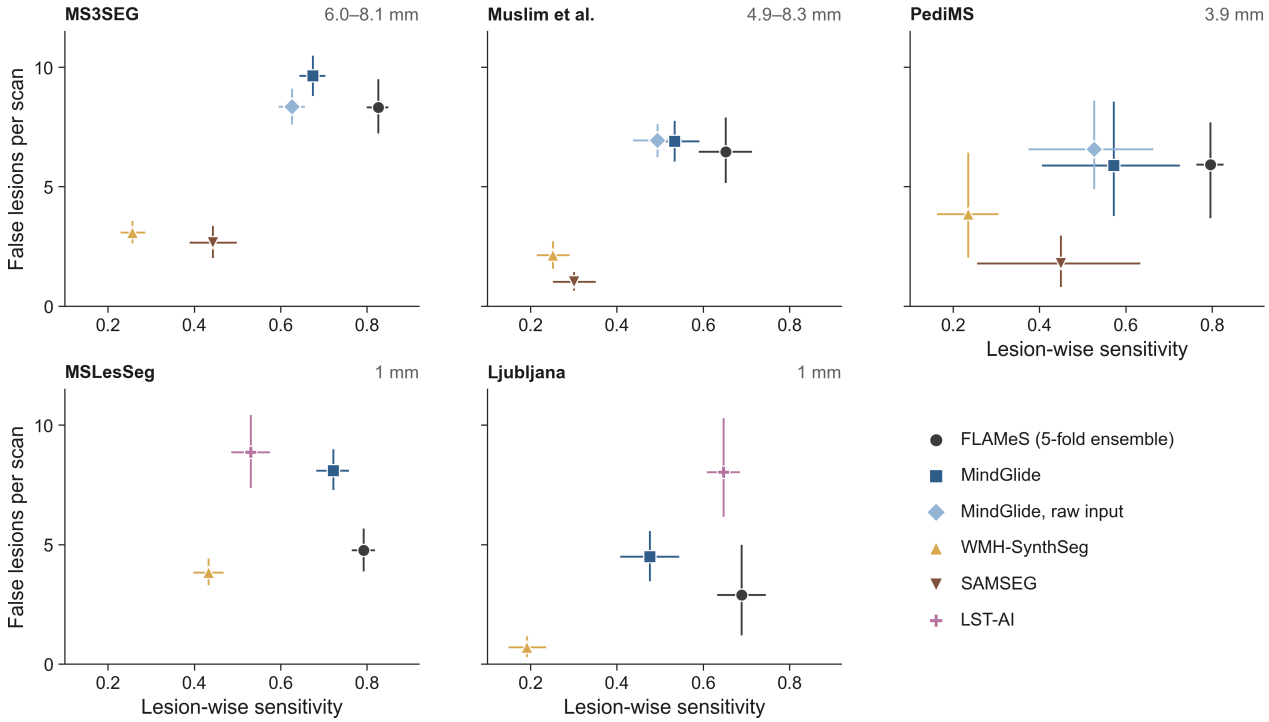


Figure 2. False lesions per scan against lesion-wise sensitivity for each tool in each cohort (mean and 95% patient-cluster bootstrap CI). The lower right is better. Top row: thick-slice cohorts; bottom row: 1 mm cohorts. On Ljubljana, FLAMeS is shown on the 10 scans not in its training and the other tools on all 30; on MS3SEG, SAMSEG covers the 50 odd-id scans (tier C).

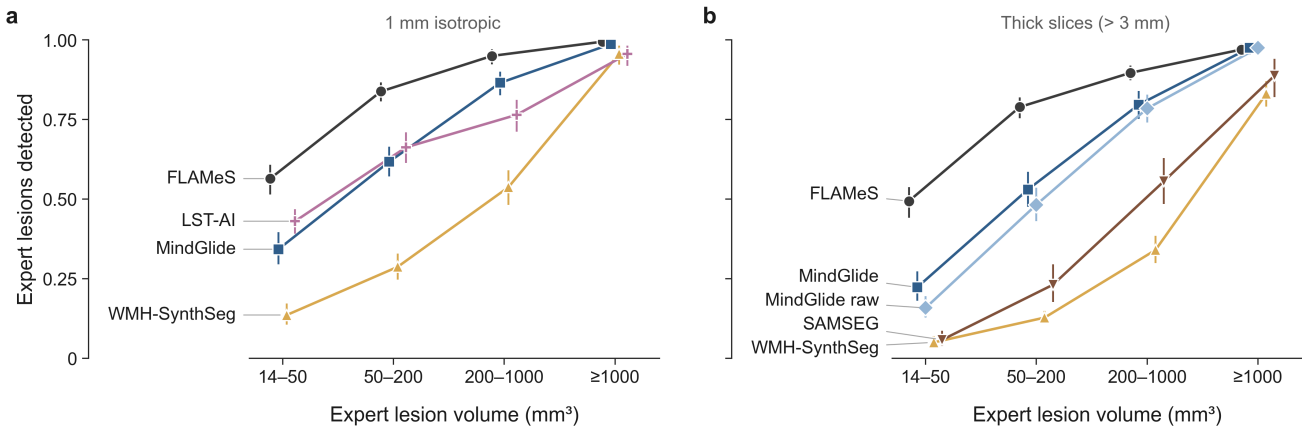


Figure 3. Proportion of expert lesions detected by lesion volume, 1 mm band (a) and thick band (b). Pooled over scans with patient-cluster bootstrap 95% CIs. SAMSEG covers the 129 tier-C scans.

3.4 Sensitivity analyses

All six thick-band sensitivity analyses (S1–S4, S6, S7) kept the sign and the 95% interval of ΔF1 below zero for all three tools (range −0.153 to −0.194 for MindGlide, −0.212 to −0.244 for SAMSEG and −0.373 to −0.436 for WMH-SynthSeg; `results/tables.md`, Table 5). The fewer false lesions of WMH-SynthSeg (−3.66 to −4.87) and SAMSEG (−5.00 to −8.25) persisted in every analysis. For MindGlide the false-lesion difference depended on the lesion definition: it became significantly positive without a size filter (S1, +2.42, 95% CI +0.10 to +5.06) and with in-plane components (S3, +8.09, +4.51 to +12.35), and stayed inconclusive in the others (for example S7, scan-weighted, +0.85, −0.34 to +2.01). Excluding Muslim (S6), whose in-plane spacing is estimated, did not change any conclusion.

Post hoc: scan-pooled F1. Per-scan F1 is unstable when a scan has few lesions (4 thick scans have fewer than 5 reference lesions, 3 of them 2 or fewer). Pooling lesions over scans (micro-averaged F1) gave the same picture: tool minus FLAMeS −0.173 (95% CI −0.202 to −0.148) for MindGlide, −0.285 (−0.352 to −0.216) for SAMSEG and −0.429 (−0.465 to −0.391) for WMH-SynthSeg on the thick band.

Post hoc: resampling of WMH-SynthSeg. WMH-SynthSeg is the only tool whose output was resampled, by the majority rule of Section 2.4. Re-scoring it from its own footprint-mean lesion probability (Table A4) changed its F1 on the thick band from 0.341 (pre-registered mask) to 0.315, 0.413 and 0.469 at probability thresholds of 0.50, 0.25 and 0.10. Its predicted volume rose from 4.55 to 13.48 mL and its detection of 14–50 mm^3 lesions from 0.050 to 0.104. Even the most permissive variant stayed far below FLAMeS (ΔF1 −0.256, 95% CI −0.309 to −0.208) and still had fewer false lesions (−5.69 per scan). The resampling rule therefore explains part of WMH-SynthSeg's under-segmentation on thick slices, but not its F1 deficit.

3.5 Secondary comparisons: the 1 mm grid and the between-band contrast

On the 1 mm-grid scans FLAMeS had F1 0.765 (0.727–0.798). It had a higher F1 than MindGlide (ΔF1 −0.272), WMH-SynthSeg (−0.363) and LST-AI (−0.180), all $p_{Holm} < 0.003$ (Table 3). MindGlide (+3.07) and LST-AI (+3.79) produced more false lesions than FLAMeS. WMH-SynthSeg produced 1.32 fewer (95% CI 0.23–2.58); this was not significant after Holm correction over the registered family ($p_{Holm} = 0.086$), although Holm over only the eight tests performed would have given 0.034. The registered rule governs. Including the 20 Ljubljana scans used in FLAMeS training (S5) gave the same conclusions (ΔF1 −0.258, −0.389 and −0.156 for MindGlide, WMH-SynthSeg and LST-AI). Because each cohort counts once, the 10 Ljubljana patients carry half of each iso estimate; per cohort, WMH-SynthSeg's ΔF1 was −0.286 on MSLesSeg (115 scans) and −0.440 on Ljubljana (10 scans), and MindGlide's −0.227 and −0.318 (post hoc;

Table A3).

FLAMeS F1 was 0.765 in the iso band and 0.724 in the thick band. MindGlide's F1 was higher in the thick band: 0.554, against 0.518 on all 145 iso scans and 0.492 on the 125 iso scans paired with FLAMeS (Figure 4a). In the registered between-band contrast, MindGlide's deficit relative to FLAMeS was 0.102 smaller in the thick cohorts than in the iso cohorts (95% CI 0.047–0.160; $p_{Holm} < 0.003$). This is statistically significant but cannot be read as smaller degradation with slice thickness: it arises because FLAMeS's F1 was lower and MindGlide's higher in the thick cohorts, and a tool whose accuracy rises when slices get thicker points to differences between cohorts, not to an effect of thickness. One such difference is MindGlide's over-segmentation on MSLesSeg (28.4 mL predicted vs 11.8 mL expert; `results/tables.md`, Table 1). For WMH-SynthSeg no difference was shown (−0.021, −0.098 to +0.056; $p_{Holm} = 1.0$). The three SAMSEG tests were not run. The iso and thick bands differ in site, scanner, population, annotation protocol and native acquisition (Section 4.3), not only in slice thickness.

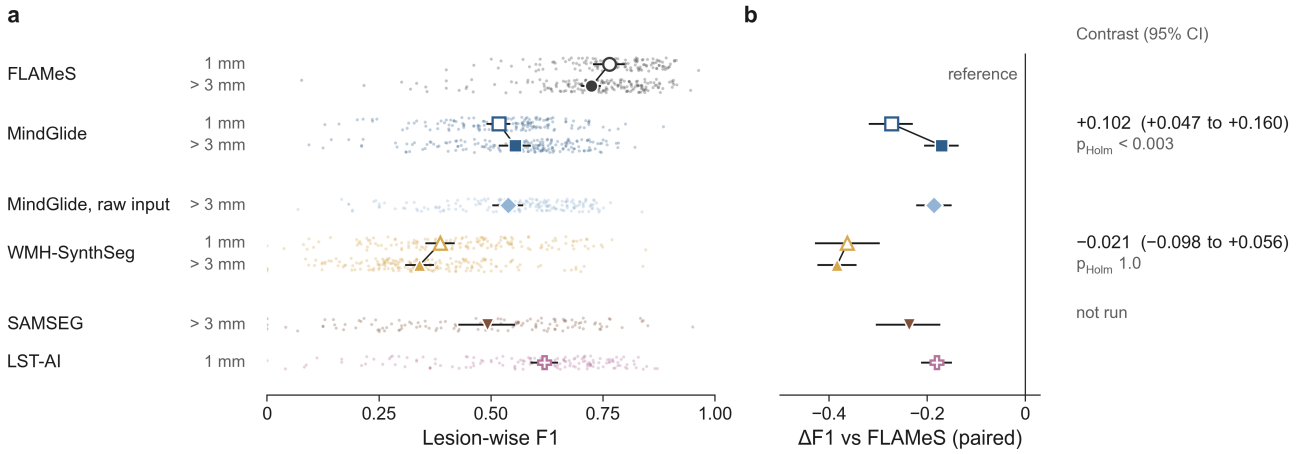


Figure 4. (a) Lesion-wise F1 per band: each dot is one scan; the marker gives the cohort-weighted mean with its 95% CI (open: 1 mm; filled: > 3 mm). FLAMeS is shown without the 20 Ljubljana scans used in its training. (b) Paired difference from FLAMeS in each band (95% CI); the slope of each line is the degradation contrast, printed at right with its 95% CI and Holm-adjusted p for the registered tests (SAMSEG: not run, because tier C is thick-only). The two bands are different cohorts, so the contrast is confounded with site.

3.6 MindGlide on the raw FLAIR

Giving MindGlide the FLAIR before skull stripping, as its documentation recommends, did not help. Against FLAMeS, raw-input MindGlide had ΔF1 −0.186 (−0.222 to −0.150; $p_{Holm} < 0.002$) and +0.38 false lesions per scan (−0.94 to +1.88; $p_{Holm} = 0.99$). Against MindGlide on the stripped input it had ΔF1 −0.016 (−0.028 to −0.004; $p_{Holm} = 0.037$) and −0.19 false lesions (−0.75 to +0.32); both differences were within the equivalence margins, so the two inputs were equivalent.

3.7 Exploratory: disagreement as a "second opinion", and a FLAMeS-only baseline

On thick scans, FLAMeS had a lesion-wise F1 below 0.5 in 14 of 179 scans (10 Muslim, 4 MS3SEG, 0 PediMS). Disagreement between MindGlide and FLAMeS identified these scans with an AUROC of 0.779; post hoc, its 95% bootstrap interval was 0.597–0.925 and the cohort-stratified AUROC was 0.819. Disagreement with WMH-SynthSeg (0.593; post hoc CI 0.385–0.787) or SAMSEG (0.488; 11 poor scans of 129) did not. On the 1 mm grid only 2 scans were poor, so the AUROCs there (MindGlide 0.813, LST-AI 0.803, WMH-SynthSeg 0.520) carry little information. Keeping only the FLAMeS lesions that a second tool also marked removed false lesions but always cost F1: with MindGlide on thick scans, −3.8 false lesions per scan (95% CI −4.7 to −2.8) and ΔF1 −0.074 (−0.111 to −0.040).

A score computed from the FLAMeS output alone did better than disagreement (post hoc, same 179 scans). Ranking scans by how few lesions FLAMeS predicted identified the poor scans with an AUROC of 0.932 (95% CI 0.861–0.987; cohort-stratified 0.956); ranking by smaller FLAMeS lesion volume gave 0.840. Combining the lesion count with MindGlide disagreement (sum of ranks) gave 0.884, lower than the count alone, and the cohort label alone (Muslim et al. or not) gave 0.733. The poor-scan label is itself fragile: three of the four poor MS3SEG scans had only 1–4 reference lesions. In these data, therefore, disagreement with MindGlide added nothing to what FLAMeS's own lesion count already showed.

3.8 Runtime and cost

Median times per scan on 4 vCPUs were 77–84 s for MindGlide (n2d-standard-4 and n2-standard-4; peak memory ≤ 5.91 GB), 253–286 s for MindGlide on raw input (e2-standard-4, an older and slower machine type; peak 13.3 GB), 216 s for WMH-SynthSeg (n2d-highmem-4; peak 31.3 GB plus up to 14.4 GB of swap) and 470–524 s per run for SAMSEG on thick scans (n2d-standard-4 and n2-standard-4; peak ≤ 7.1 GB, pilot included); all met the pre-stated CPU rule of ≤ 15 minutes per scan. For comparison, FLAMeS took 67.5–69.1 s per fold and scan on the same two machine types (about 5.7 minutes for the five folds without test-time mirroring). Processor families differed (AMD Milan, Intel Cascade Lake, Intel Broadwell), so runtimes are indicative. Table 5 gives the cost.

Table 5. Compute for this study (list prices; `data/compute_ledger_c8b4239.csv`).

Component	Runs	Machine-hours	USD
Pilots (MindGlide, FreeSurfer tools)	5	1.54	0.30
MindGlide, primary arm (324 scans)	2	6.78	1.32
MindGlide, raw-input arm (179 scans)	2	13.07	1.93
WMH-SynthSeg (324 scans, incl. one `--crop` re-run)	2	22.65	5.63
SAMSEG, tier C (127 scans + 2 from pilots)	3	17.86	3.37
Total, this study	**14**	**61.90**	**12.55**
Earlier runs reused: FLAMeS maps, 1 mm cohorts	3	14.24	2.62
Earlier runs reused: HD-BET + FLAMeS maps, thick cohorts	2	21.79	3.83
Earlier runs reused: LST-AI maps	1	4.05	0.84
Total including reused runs	**20**	**101.97**	**19.84**

With the real per-scan cost of SAMSEG, tier B (all 179 thick scans) would have fitted under the cap (about 13.9 USD in total) but tier A would not (about 17.8 USD) (`results/RESULTS.md`, Section 5).

4. Discussion

4.1 What the comparisons show

The registered questions were two-sided, and the answer is clear. On real thick-slice FLAIR, FLAMeS had a significantly higher lesion-wise F1 than each public CPU tool; SAMSEG and WMH-SynthSeg had significantly fewer false lesions; MindGlide's false-lesion count did not differ detectably. No tool met the reading rule for outperforming

FLAMeS, and none was equivalent to it. The F1 deficits (0.17–0.38 F1) are several times the ±0.04 precision the protocol anticipated, every 95% interval lay below the margin of −0.05, and every cohort showed the same direction, although FLAMeS's training data, as recorded in its released fingerprint, contained no voxel dimension larger than 1.25 mm. The data do not say that these tools are poor for their intended purposes: WMH-SynthSeg targets white-matter hyperintensities of any cause in heterogeneous and low-field scans (Laso et al. 2024), SAMSEG provides whole-brain anatomy with lesions (Cerri et al. 2021), and MindGlide extracts brain and lesion volumes from clinical archives for research (Goebl et al. 2025). Lesion-by-lesion detection against expert masks at default settings is a stricter, different task.

The conditions matter: identical skull-stripped input, default settings and binarisations, 3D components ≥ 14.1 mm^3 matched by any overlap, and each cohort's own reference protocol. The direction of the F1 differences was robust to the lesion definition (S1–S3), weighting (S7), the cohort with estimated geometry (S6), failure handling (S4), pooling across scans and, for WMH-SynthSeg, the resampling rule; for MindGlide, skull stripping was not the cause. The matching rule needs its own caution: any-overlap matching of 26-connected components across 6–8 mm slices rewards large merged components, which favours over-segmenting tools on false-lesion counts, whereas the MSSEG-2016 evaluation required a minimum overlap and limited how far a detecting component may extend beyond the reference lesion (Commowick et al. 2018). Our false-lesion outcome is fragile in this way: MindGlide's difference moved from +0.57 to +8.09 when components could not join across slices (S3), while the F1 ranking did not change.

4.2 Fewer false lesions are not a better operating point

SAMSEG and WMH-SynthSeg produced significantly fewer false lesions than FLAMeS, which in isolation might suggest that they suit thick-slice archives better. Figures 2 and 3 show why not: both mark a fraction of the expert lesion volume, detect only 5–6% of small expert lesions and miss most lesions below 200 mm^3. Their precision is similar to or higher than that of FLAMeS but their recall is far lower; fewer false lesions are the price of under-segmentation. Other probability thresholds would move them along their precision–recall curves, which this study, keeping defaults, did not explore beyond the post hoc WMH-SynthSeg analysis. The MindGlide paper reported that WMH-SynthSeg failed on 15% of visually assessed contrasts, particularly with slices thicker than 5 mm (Goebl et al. 2025). We saw no outright failures, but on thick slices WMH-SynthSeg marked only about a quarter to a third of the expert lesion volume in every cohort. Part of this is due to our majority rule, which removes thin lesions; scoring its own probabilities brought its volume close to the expert's, yet its F1 stayed far below that of FLAMeS (Section 3.4). Part of WMH-SynthSeg's apparent false-lesion count on MS3SEG may also come from the dataset's exclusion of "normal" periventricular hyperintensities from the reference.

4.3 Contamination and site confounding

Training overlap. FLAMeS was trained on ISBI-2015, MSSEG-2016 and 20 Ljubljana scans (Dereskewicz, La Rosa, Dos Santos Silva, et al. 2025); we identified the 20 from the released fingerprint and excluded them, and including them (S5) changed no conclusion. ISBI-2015 and MSSEG-2016 are not among our cohorts, but they shaped WMH-SynthSeg (training label maps) and SAMSEG (evaluation), and Ljubljana served as an external test set of MindGlide and LST-AI; these overlaps concern development history, not the scans evaluated here. The FLAMeS developers reported results on 51 MSLesSeg cases before release; if that influenced model selection, FLAMeS may be slightly favoured on the iso band, but not in the thick-band primary comparison. The thick cohorts (MS3SEG data 2025, article 2026) were all public before the FLAMeS v2.0 weights, yet no FLAMeS training case has a voxel dimension above 1.25 mm (Section 2.3).

Site confounding. The primary comparisons are paired within scan, so scanner, population and annotation protocol affect FLAMeS and the comparator alike, and the cohort weighting prevents the largest cohort from dominat-

ing. Two analyses are not protected. First, the between-band ("degradation") contrast compares different cohorts, which differ in field strength, vendor, age group, skull stripping, rater protocol and native acquisition: 115 of its 125 FLAMeS comparison scans come from MSLesSeg, acquired with slices of about 2.6 mm and resampled to 1 mm, and 10 Ljubljana patients carry half of every iso estimate. MindGlide's F1 was higher in the thick cohorts, which slice thickness cannot explain; its 2.4-fold over-segmentation of lesion volume on MSLesSeg (28.4 mL predicted vs 11.8 mL expert) is one cohort-specific cause. The contrast is therefore not evidence that any tool tolerates thick slices better than FLAMeS; a within-subject design, with the same patients scanned or resampled at both resolutions, would be needed. Second, cohort-level differences in reference standards (for example MS3SEG's exclusion of normal hyperintensities and the sequential correction chain in Muslim et al.) affect absolute values and cross-cohort comparisons. They do not explain away paired differences of the size observed.

4.4 The exploratory disagreement signal does not survive a simple baseline

Disagreement between MindGlide and FLAMeS ranked the thick scans on which FLAMeS performed poorly with an AUROC of 0.78 (exploratory; post hoc 95% interval about 0.60 to 0.93). A score that needs no second model did better: ranking scans by how few lesions FLAMeS itself predicted gave an AUROC of 0.93 (post hoc), and adding the disagreement score did not improve it. In these data the disagreement score therefore mostly tracks "FLAMeS marked few lesions". Both analyses rest on 14 poor scans, 10 of them from one cohort, and the poor-scan label is unstable for scans with very few reference lesions; using a second tool as a lesion filter also always cost F1. We therefore do not propose a "second-opinion" flag on the basis of this study. Any quality-control rule, including one based on FLAMeS's own lesion count, would need its own pre-specified threshold and validation on independent data. We make no claim about clinical value.

4.5 Relation to previous work and implications

The FLAMeS developers compared FLAMeS with SAMSEG, LST-LPA and LST-AI and reported that it outperformed them on their external test sets, which included a small clinical cohort with 2D anisotropic FLAIR (Dereskewicz, La Rosa, Dos Santos Silva, et al. 2025); our 1 mm-grid results agree in direction, and our results extend the comparison to 179 thick-slice scans from three further sources. LST-AI, which uses a T1 as well, had a lower F1 and more false lesions than FLAMeS on the 1 mm grid. To our knowledge, this is the first pre-registered, paired comparison on real thick-slice FLAIR from several sources with expert masks. For researchers working with thick-slice archives, the results suggest three practical points: (i) report lesion-wise metrics stratified by slice thickness and lesion size, not only Dice or total volume; (ii) treat low false-lesion counts with suspicion until sensitivity is reported alongside them; and (iii) where a CPU is the constraint, a 5-fold nnU-Net ensemble ran in about 5.7 minutes per scan on 4 vCPUs here, so a GPU is not required to use it. None of this establishes clinical utility, which would need clinical end points, reader studies and prospective data.

5. Limitations

Data and inputs. SAMSEG covers 129 of the 179 thick scans and no 1 mm-grid scan; tier B would have fitted the budget, and completing it on the other 50 MS3SEG scans would need new cloud compute. All tools received skull-stripped input; SAMSEG's atlas registration prefers images with skull, and only MindGlide was also tested on raw input. The reference standards are single masks made under different protocols, the iso band is mostly MSLesSeg resampled from ~2.6 mm slices, Muslim et al. pixel sizes are estimated, and the data are research-grade public scans; no clinical outcome was studied.

Analysis. Default operating points differ between tools; only WMH-SynthSeg's output was resampled; any-overlap

matching across thick slices makes the false-lesion outcome fragile (S3); the cohort weighting gives small cohorts large weight; the between-band contrast cannot isolate slice thickness; and the exploratory analyses rest on 14 poor scans. The Results and Discussion report how each of these affects the findings.

Process. The protocol was time-stamped in a public git repository, not an independent registry. One author, whose software uses the reference model, designed and analysed the study with an AI coding agent, and no independent statistician audited it [TODO owner: consider an independent co-author or statistical audit]. An unplanned interim analysis was run, and the timing of the structural check rests on a session log kept outside the repository (Table 6). The nnU-Net version of the FLAMeS runs was not recorded in the manifests [TODO owner].

6. Conclusion

In pre-registered, paired comparisons on 179 real thick-slice FLAIR scans from three public cohorts, the FLAMeS 5-fold ensemble had a substantially higher lesion-wise F1 than MindGlide, WMH-SynthSeg and SAMSEG, and no tool met the pre-specified criterion for outperforming it. The two FreeSurfer tools produced fewer false lesions only by missing most small lesions. On the 1 mm grid FLAMeS also had the highest F1 of the tools tested there, including LST-AI (SAMSEG was not run on them). Neither the between-band contrast nor the exploratory disagreement score gave an interpretable signal: the first mixes cohorts, and the second was outperformed by FLAMeS's own lesion count. The protocol, every per-scan metric and the code are public so that others can test other tools under the same rules.

Pre-registration deviations

Table 6. Deviations and notable events (protocol Section 12, plus one item identified while writing this manuscript). "Seen before" lists what had been observed when the change was made.

Time (UTC)	Change	Reason	Seen before
2026-10-21 20:55 (commit; first push 20:58)	Wording of the MindGlide command description (its cleanup step acts on the whole label map)	clarification; no rule changed	nothing launched
2026-10-21 22:43 (commit and push)	Secondary arm 7.6 registered: MindGlide on the raw FLAIR (thick band), own Holm family of 4	MindGlide's documentation asks for raw input	MindGlide pilot only (3 scans: timings, one lesion volume). The MindGlide full-run machines had started at 22:32–22:34 and were computing, so some full-run outputs probably existed; none had been examined
2026-10-22 20:08 (push of 6d507c4)	**Unplanned interim analysis** of MindGlide (both arms) and LST-AI vs FLAMeS, although the protocol stated a single analysis on full outputs; not logged at the time, identified while writing this manuscript	progress report while the FreeSurfer tools awaited a licence	interim results for MindGlide, LST-AI and FLAMeS on all cohorts, including the exploratory E1 AUROC of 0.78, which the product-planning note then read as meeting its "second-opinion" criterion; the final analysis reproduces the interim numbers exactly. All later decisions below were taken after this interim, but before any WMH-SynthSeg or SAMSEG accuracy output

Time (UTC)	Change	Reason	Seen before
2026-10-24 19:37 (push)	WMH-SynthSeg run without `--crop` although the pilot peak (31.32 GB) exceeded the pre-stated 30 GB trigger; kept for all scans; one out-of-memory scan re-run with `--crop`. The decision was taken by the AI agent acting as run coordinator, which applied a conflicting runbook rule	conflicting rule in the runbook; uniform processing preferred	runtime and memory only
2026-10-24 19:37 and 2026-10-28 14:54 (pushes)	Budget rule (pre-stated) first dropped SAMSEG on projected cost, then, re-run with the real spend, selected tier C (129 scans); boot disk 80 GB instead of 150 GB	application of the pre-stated rule	runtime and cost only
2026-10-29 00:01 (push of 7a9a24e)	SAMSEG analysis set = tier C; its one 1 mm pilot output enters no comparison; three SAMSEG secondary tests not run	a single 1 mm scan gives a one-cluster bootstrap	run-integrity checks only
2026-10-29 00:01 (same push)	Holm over the registered family size, tests not run entered as p = 1	the secondary family could not be complete	run-integrity checks only
2026-10-29 00:01:59 (session log; not in the repository)	Structural check of the WMH-SynthSeg and SAMSEG outputs, which also computes per-scan Dice, lesion-wise F1 and volume (Section 2.9)	planned check before the analysis	— (run 21 s after the push above and 9 min before the final analysis at 00:11)
2026-10-29 (note)	One SAMSEG pilot output (ms3seg_001) lacks a manifest record; verified by voxel count and kept	pilot manifest write crashed (bug fixed)	voxel count only
2026-10-31	Post hoc, labelled additions in this manuscript: E1 interval, stratified AUROC and FLAMeS-only baselines; precision, recall and Dice per band; FLAMeS on the SAMSEG set; paired-set absolute F1; per-cohort differences; scan-pooled F1; WMH-SynthSeg resampling variants (fixed in the script before computing); fingerprint check	interpretation; internal review	all results

Declarations

Data availability

All five cohorts are public (Table 7). MSLesSeg, MS3SEG, Muslim et al. and PediMS are distributed under CC BY 4.0. Ljubljana is distributed under CC BY-NC-ND according to its laboratory's page (a GitHub mirror states CC BY); we follow the stricter licence, so no mask derived from Ljubljana will be deposited. The per-scan metrics of every tool (`research/paper_f/results/per_scan.csv.gz`, 6,906 rows), all comparisons, the run-integrity report, the compute

ledger snapshot and the numbers added for this manuscript are in the public repository https://github.com/nicolasbonilla/medical-imaging-viewer (branch `research/paper-f-tool-benchmark`, results commit `1d3be48`; manuscript branch `research/paper-f-manuscript`). The SHA-256 of every input file is in `run_info.json`. The tool output masks are kept in the authors' research storage [TODO owner: decide whether to deposit the derived masks of MSLesSeg, MS3SEG, Muslim et al. and PediMS, e.g. on Zenodo with a DOI, respecting the dataset and tool licences; Ljubljana-derived masks are excluded (ND), and the MS3SEG and Muslim inputs were skull-stripped with non-commercial HD-BET parameters]. Every number in this manuscript is mapped to its source file and field in `NUMBERS_CHECK.md`.

Code availability

Analysis code (`metrics.py`, `evaluate.py`, `summarize.py`, `verify_runs.py`, `plan_budget.py`), the cloud job scripts (`research/cloud/`), the figure script (`make_figures.py`) and the post hoc scripts (`compute_extra_numbers.py`, `wmh_resampling_posthoc.py`, `fingerprint_check.py`) are in the same repository; local tests are in `research/paper_f/tests/`. [TODO owner: the repository has no licence file; add one (for example MIT for the research code) and archive a tagged release with a DOI before submission.]

Ethics statement

This study used only public, de-identified datasets released by their authors under open licences, with ethics approval and consent handled by the original studies as described in their data papers. No new data were collected and no participant was contacted. [TODO owner: confirm whether the journal or the author's institution requires a formal waiver statement for secondary analysis of public data.]

Licences

Table 7. Licences of data and software as used here (checked 2026-10-21, 2026-10-28 and 2026-10-31).

Item	Licence	Consequence for this study
MSLesSeg data	CC BY 4.0	attribution; changes stated
Ljubljana (Lesjak et al. 2018)	CC BY-NC-ND according to the laboratory's page (checked 2026-10-28); the open_ms_data GitHub mirror states CC BY in its README and has no licence file	we follow the stricter licence: academic research only; no derived masks deposited (ND); no use for product calibration, thresholds or marketing (NC)
MS3SEG data	CC BY 4.0 (article text CC BY-NC-ND 4.0)	attribution; changes stated
Muslim et al. data	CC BY 4.0	attribution; reorientation and estimated pixel size stated
PediMS data	CC BY 4.0	attribution
FLAMeS weights	CC BY 4.0; nnU-Net code Apache-2.0	attribution
LST-AI	MIT; documented as research-only and not approved for clinical use	research use
MindGlide	code MIT; no licence stated for the weights	research use; ask the authors before any other use
FreeSurfer 8.2.0 (WMH-SynthSeg, SAMSEG)	FreeSurfer Software License Agreement (registration); states research purposes, clinical applications not advised	research use

Item	Licence	Consequence for this study
HD-BET 2.0.1	code Apache-2.0; model parameters CC BY-NC 4.0 (Zenodo record 14445620, downloaded by HD-BET 2.x) (Isensee et al. 2024)	used to skull-strip 151 of the 179 thick scans (MS3SEG, Muslim et al.): non-commercial use only. The thick-band results rest on these inputs, so neither the stripped images nor numbers derived from them should be used for commercial purposes without a separate licence or a different skull stripper

Conflict of interest

Nicolás Bonilla Vargas (Universidad Nacional de Colombia; SRH University München; Daita AI) develops MSTool-AI, investigational software for MRI in multiple sclerosis. MSTool-AI's research pipeline already uses FLAMeS, the reference model of this study, so a result in favour of FLAMeS supports a choice the author has already made for his software: the result favours the author's commercial interest. The product-planning note committed with the protocol (`docs/research/paper-f/APP-FEATURE.md`) pre-specified product features that would use this paper's numbers: a thick-slice "reliability notice" quoting the F1 and false-lesion figures reported here, and an exploratory "second-opinion" flag based on analysis E1, which this study no longer supports (Section 4.4). The reference model was chosen by the author, and the same author, assisted by an AI coding agent, designed, ran and analysed every arm; no independent statistician audited the analysis. The author has no financial relationship with the developers of any tool tested. [TODO owner: confirm this statement; declare any role or financial interest in Daita AI or SRH University relevant to this work; add co-authors' declarations.]

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This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors. Cloud compute was paid from the promotional free-trial credit of the author's Google Cloud account. Role of the author's company interests: none beyond those stated under Conflict of interest; the planned product use of the results is disclosed there.

Author contributions (CRediT)

Nicolás Bonilla Vargas: Conceptualization, Project administration, Resources, Writing – review & editing [owner to confirm, and to add any other roles he performed]. Software, formal analysis, visualisation and the original draft were produced by an AI coding agent (Section 2.1 and the declaration below); CRediT roles apply to human authors only. [TODO owner: co-authors and their roles.]

Acknowledgements

We thank the creators of MSLesSeg, the Ljubljana dataset, MS3SEG, the Muslim et al. dataset and PediMS for releasing their data, and the developers of FLAMeS, nnU-Net, LST-AI, MindGlide, FreeSurfer, WMH-SynthSeg, SAMSEG, HD-BET and SynthStrip for releasing their software.

Appendix

Table A1. Software identity.

Component	Identifier
MindGlide 1.3.0 container	`ghcr.io/ms-pinpoint/mindglide@sha256:14e88f61d6cba18322d03774b168682e255383387aaab55e6a5aa33a2dbdd971`
MindGlide weights	`_20240404_conjurer_trained_dice_7733.pt`, SHA-256 `881e30efd9444a25ee70c01d795dd9fb21ac750a48f1ba8070fcd79fb75e76ca`
FreeSurfer 8.2.0 container	`freesurfer/freesurfer@sha256:5648104ca75a24f6ee5a8786146f99b405c2bda1da12bff723a15056da8ddf6b` (build `freesurfer-linux-rocky8_x86_64-8.2.0-20260314-d932c45`)
WMH-SynthSeg model	`WMH-SynthSeg_v10_231110.pth`, 790,531,383 bytes, SHA-256 `0ece39dd651357aa95222fc4d45fa32d00f11e763d2583cae3f869989ce35988`
LST-AI 1.2.0 container	`jqmcginnis/lst-ai@sha256:2b2a0d6454e6efff60a350df3c650ae9584f1926338dec7842f9e71912cc6c85`
FLAMeS weights	Zenodo 10.5281/zenodo.17955359, version 2.0 (`Dataset004_WML.zip`)
Protocol at registration	commit `65a95e3`; `PROTOCOL.md` SHA-256 `587ba9ff…f284e2`; `metrics.py` `8aae0587…5637f`; `evaluate.py` `93512380…f3032`; `flames_inputs.json` `4847df6d…e398d`
Final analysis	commit `7a9a24e`; input-list digest `d50d356bcd6676b38fb10545f1f0a659a49582ba4d83ad8c5567bdcc9dd0b43b`

Table A2. Run integrity (`results/verify_runs.json`).

Tool	Planned	Present, hash = machine record	Never recovered	Failed, then re-run
WMH-SynthSeg	324	324	0	1 (out of memory → `--crop`)
MindGlide, primary	324	324	0	3 (pilot setup error)
MindGlide, raw input	179	179	0	3 (10 h time guard)
SAMSEG, tier C	129	128 + 1 verified by voxel count	0	0

Appendix B. Training-overlap checks against the released FLAMeS weights. The FLAMeS paper lists 20 Ljubljana scans in training without naming them (Dereskewicz, La Rosa, Dos Santos Silva, et al. 2025). The released weights include nnU-Net's dataset fingerprint (per-case spacing and cropped brain size of all 668 training cases). Twenty consecutive entries matched the bounding box of exactly one Ljubljana patient each with zero voxels of difference (patients 01–15 and 21–25), leaving patients 16–20 and 26–30 never seen by any fold (`data/flames_ljubljana_split_87e998d.json`). These 20 scans were excluded from every comparison with FLAMeS and appear only in sensitivity analysis S5. MSLesSeg was not in FLAMeS training; the FLAMeS developers used 51 MSLesSeg cases as an external test set. The same fingerprint gives two further checks. None of the 115 MSLesSeg scans matches the bounding box of any of the 319 training entries with 1 mm spacing (`data/mslesseg_flames_overlap_check_87e998d.json`). None of the 668 training entries has a voxel dimension above 1.25 mm (`data/flames_fingerprint_check.json`), so no thick-slice scan of MS3SEG, Muslim et al. or PediMS was a FLAMeS training case at its own geometry; a resampled copy cannot be excluded by this check, although the three cohorts come from centres not listed among the training sources of FLAMeS, MindGlide, WMH-SynthSeg, SAMSEG or LST-AI, and all three were public before the FLAMeS v2.0 weights (2025-12-16): Muslim et al. in 2022, PediMS in May 2025 and the MS3SEG data in October 2025. The fingerprint also shows that FLAMeS never saw a voxel dimension larger than 1.25 mm in training.

Table A3. Per-cohort paired differences from FLAMeS (post hoc; tool − FLAMeS, mean over scans with a single-cohort patient bootstrap; `data/extra_numbers.json`, `revision.per_cohort_paired_differences`).

Tool	Cohort	Scans	ΔF1 (95% CI)	Δ false lesions / scan
MindGlide	MS3SEG	100	−0.162 (−0.182 to −0.143)	+1.31
MindGlide	Muslim et al.	51	−0.123 (−0.157 to −0.089)	+0.43
MindGlide	PediMS	28	−0.226 (−0.324 to −0.127)	−0.04
MindGlide	MSLesSeg	115	−0.227 (−0.256 to −0.200)	+3.33
MindGlide	Ljubljana	10	−0.318 (−0.408 to −0.235)	+2.80
WMH-SynthSeg	MS3SEG	100	−0.404 (−0.436 to −0.371)	−5.24
WMH-SynthSeg	Muslim et al.	51	−0.279 (−0.333 to −0.224)	−4.33
WMH-SynthSeg	PediMS	28	−0.467 (−0.570 to −0.368)	−2.07
WMH-SynthSeg	MSLesSeg	115	−0.286 (−0.324 to −0.250)	−0.93
WMH-SynthSeg	Ljubljana	10	−0.440 (−0.567 to −0.313)	−1.70
SAMSEG	MS3SEG (odd ids)	50	−0.229 (−0.275 to −0.186)	−5.86
SAMSEG	Muslim et al.	51	−0.226 (−0.288 to −0.165)	−5.45
SAMSEG	PediMS	28	−0.255 (−0.444 to −0.077)	−4.14
LST-AI	MSLesSeg	114	−0.214 (−0.253 to −0.177)	+4.08
LST-AI	Ljubljana	10	−0.147 (−0.199 to −0.098)	+3.50

Table A4. WMH-SynthSeg on the thick band under the pre-registered resampling and three post hoc variants that threshold its footprint-mean lesion probability (179 scans; cohort-weighted; ΔF1 vs FLAMeS with 95% CI; `data/wmh_resampling_posthoc.json`).

Variant	Lesion-wise F1	ΔF1 vs FLAMeS (95% CI)	Δ false lesions / scan	Recall	Predicted mL	14–50 mm^3 detected
Pre-registered: footprint majority of the hard label	0.341	−0.384 (−0.424 to −0.344)	−3.88	0.248	4.55	0.050
Probability ≥ 0.50	0.315	−0.409 (−0.451 to −0.367)	−4.28	0.224	4.08	0.044
Probability ≥ 0.25	0.413	−0.312 (−0.360 to −0.266)	−5.17	0.304	8.49	0.072
Probability ≥ 0.10	0.469	−0.256 (−0.309 to −0.208)	−5.69	0.352	13.48	0.104

Supplementary tables (per cohort, sub-bands, sensitivity analyses, size bins, exploratory analyses) are `research/paper_f/results/tables.md`, generated by `summarize.py` from the pre-registered outputs.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used an AI coding agent (Claude Opus 5.5, Anthropic, used through Claude Code) to verify the references, to compute and check every number in the text, and to write the first draft of the manuscript and its revision; its role in the research itself (protocol drafting, all code, run-time decisions and the analyses) is described in Methods, Section 2.1. [Owner to confirm: "After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication." This sentence must be true before submission and must be confirmed by the author himself.] The agent is not an author. No generative AI system created or altered any image or mask: all masks were produced by the named segmentation tools, which are themselves AI models, and all measurements by the committed scripts.

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